



GS FOUNDATION 2024-25

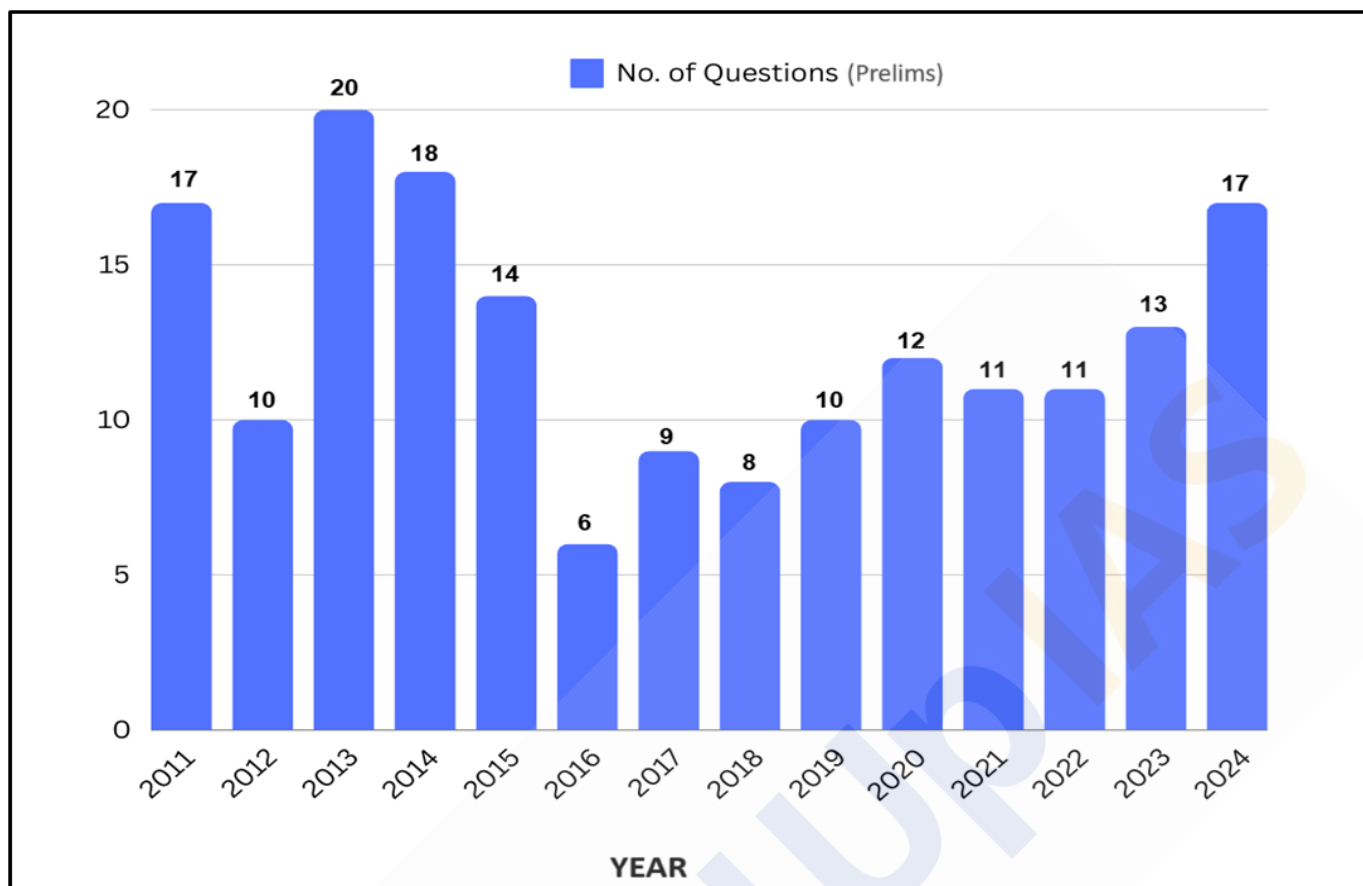
GEOGRAPHY - 4

Atmospheric Pressure and Winds

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Atmospheric Pressure and Winds

Introduction:

We do not ordinarily think of air as having too much weight. But **air has weight and it exerts pressure**. But we do not feel the weight of the atmosphere because we have air inside us which exerts an equal outward pressure that balances the inward pressure of the atmosphere. Atmospheric pressure is important to us because it is related to winds and it helps to determine weather conditions of a place.

Atmospheric Pressure:

The atmospheric pressure is the **weight of a column of air per unit area extending from the mean sea level to the top of the atmosphere**. This pressure is measured using a device called a **barometer**.

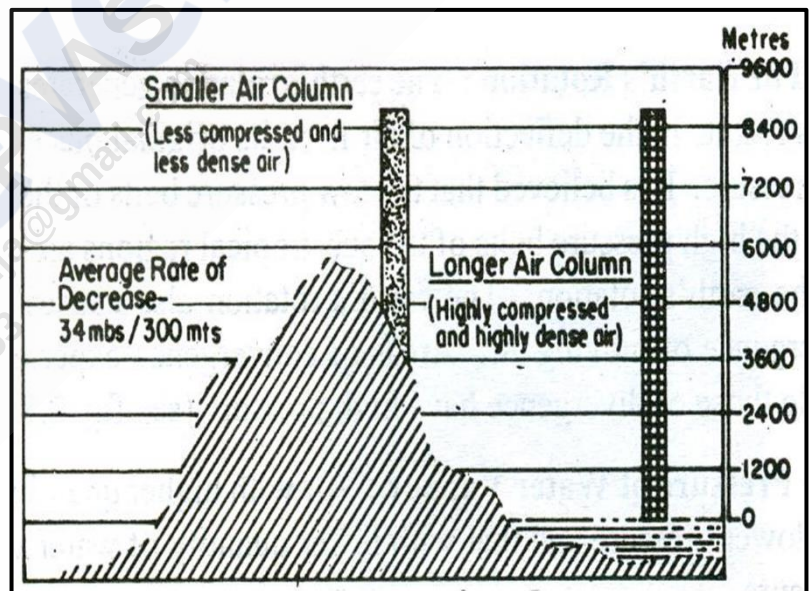
Atmospheric pressure is quantified as **force per unit area, with millibars (mb)** being the standard unit of measurement. One millibar approximates the force exerted by one gram per square centimetre. **Distribution of atmospheric pressure** on the surface of the earth is not uniform. It varies both **vertically and horizontally**.

Vertical Distribution of Pressure:

Air is a mixture of various gases and is highly compressible. As air compresses, its density increases, leading to higher air pressure. **The density of air is greater in the lower layers of the atmosphere** due to the mass of air above compressing it. Consequently, lower atmospheric layers have higher density and **exert more pressure, while the upper layers are less compressed, with lower density and pressure**.

Factors Influencing Vertical Distribution:

The vertical distribution of atmospheric pressure refers to how pressure changes with altitude. Generally, **air pressure decreases with an increase in altitude, but this decrease is not always consistent**. The densest components of the atmosphere are found near the mean sea level. **Factors such as air temperature, water vapour content, and Earth's gravitational pull** determine air pressure at a given place and time. Since these factors vary with height, the rate of decrease in air pressure with altitude also varies. On average, **air pressure decreases by 34 millibars for every 300 meters of altitude gain**.



Additionally, **rising atmospheric pressure** typically indicates **fine, settled weather**, whereas **falling pressure** suggests **unstable and cloudy conditions**.

Effects of Low Pressure in High Altitudes:

People living in hilly and mountainous areas experience the effects of low pressure more acutely than those in plains. **In high mountainous regions, low pressure lowers the boiling point of water, causing rice and other foods to take longer to cook**. Trekkers and visitors often face **breathing problems**, such as faintness and nosebleeds, due to the thin air and low oxygen content at higher altitudes.

Horizontal Distribution of Pressure:

The horizontal distribution of atmospheric pressure over the globe is represented on maps using 'isobars', which are **lines connecting points of equal pressure**. Isobars are similar to contour lines on relief maps, and their spacing indicates the rate and direction of pressure change, known as the pressure gradient. **A close spacing of isobars signifies a steep pressure gradient, while a wide spacing indicates a gentle pressure gradient.** The horizontal distribution of atmospheric pressure is not uniform; it varies from time to time at a given place and from place to place over short distances.

Factors Influencing Horizontal Distribution:

Air Temperature:

- We have studied that the earth is not heated uniformly because of unequal distribution of insolation, differential heating and cooling of land and water surfaces. Generally, there exists an **inverse relationship between temperature and pressure** - higher temperatures correspond to lower pressure. This is because **heated air becomes less dense, expands, and rises, creating low-pressure areas**. Consequently, equatorial regions experience low pressure due to the ascent of heated air, while polar regions exhibit high pressure as cold, dense air descends. However, pressure does not increase uniformly from the equator to the poles. Instead, there are high-pressure regions in the subtropics and low-pressure regions in the subpolar areas.

The Earth's Rotation:

- The Earth's rotation generates **centrifugal force, which deflects air from its original path and decreases pressure**. This rotation is believed to create the low-pressure belts in the subpolar regions and the high-pressure belts in the subtropical regions. Additionally, the Earth's rotation **causes convergence and divergence** of moving air, with **convergence areas experiencing low pressure** and **divergence areas experiencing high pressure**.

Presence of Water Vapour:

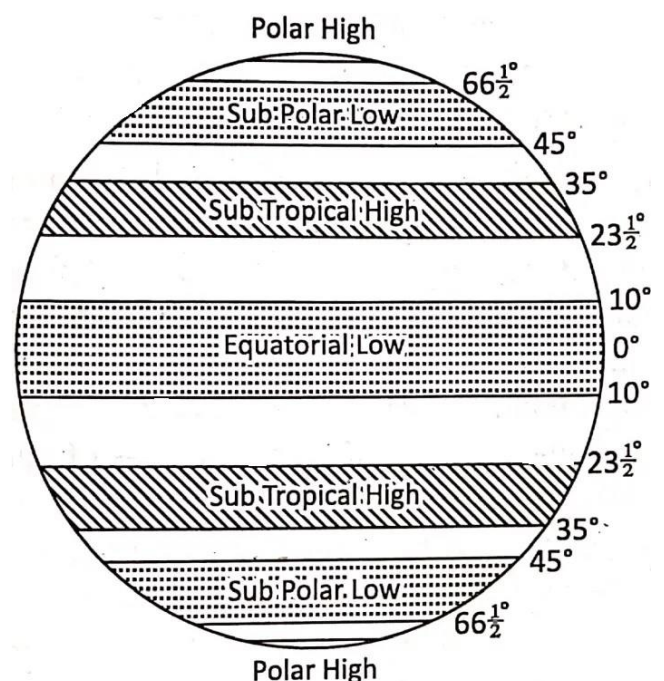
- Air with a higher quantity of water vapor has lower pressure**, while **air with less water vapor has higher pressure**. During winter, continents are relatively cool and tend to develop **high-pressure centers**. In summer, continents are warmer than oceans and are dominated by **low pressure**. Conversely, oceans exhibit low pressure in winter and high pressure in summer.

Earth's Pressure Belts:

Understanding the distribution of air pressure across Earth's latitudes is crucial for comprehending global weather patterns and climate dynamics. The concept of pressure belts, ranging from equatorial lows to polar highs, forms a fundamental framework in meteorology. These belts, though theoretical in nature, provide insights into how solar heating, Earth's rotation, and atmospheric circulation interact to shape regional and global weather phenomena.

These pressure belts are:

- The Equatorial Low-Pressure Belt
- The Sub-tropical High-Pressure Belts
- The Sub-polar Low-Pressure Belts
- The Polar High-Pressure Belts



Equatorial Low-Pressure Belt:→ Benonic

- The **sun shines almost vertically on the equator** throughout the year. As a result, the air gets warm and rises over the equatorial region and produce equatorial low pressure. This belt extends from **equator to approximately 10° North and South latitudes**.
- Due to excessive heating, horizontal movement of air is minimal here and only **conventional currents** are there. Therefore, this belt is called **doldrums** (the zone of calm) due to virtual absence of surface winds. These are the regions of convergence because the **winds flowing from sub-tropical high-pressure belts converge** here. This belt is also known as **Inter Tropical Convergence Zone (ITCZ)**.

Sub-tropical High-Pressure Belts

→ cold wind set down making High Pres. zone

- Extending from the **tropics to around 35° latitudes in both hemispheres**. In the northern hemisphere it is called as the **North sub-tropical high-pressure belt** and in the southern hemisphere it is known as the **South sub-tropical high-pressure belt**.
- The existence of these pressure belts is due to the fact that the **rising air of the equatorial region** is **deflected towards poles** due to the earth's rotation. After **becoming cold and heavy**, it **descends** in these regions and gets piled up. This results in **high pressure**.
- **Calm conditions** with feeble and variable winds are found here. In olden days vessels with cargo of horses passing through these belts found **difficulty in sailing under these calm conditions**. They used to **throw the horses in the sea** in order to make the vessels lighter. Henceforth these belts or latitudes are also called '**Horse latitudes**.' These are the regions of divergence because winds from these areas blow towards equatorial and sub polar low-pressure belts.

Sub-polar Low-Pressure Belts:

- The Sub-polar Low-Pressure Belts are situated between approximately **45°N and the Arctic Circle in the northern hemisphere**, and **45°S and the Antarctic Circle in the southern hemisphere**.
- These regions are characterized by the convergence of polar easterlies from the polar high-pressure belts and westerlies from the subtropical high-pressure belts.
- This convergence often leads to the formation of cyclonic storms and frontal systems known as the polar front, which play a crucial role in global weather patterns.

Polar High-Pressure Belts:

- In polar regions, **between 80° to 90° North and South**, sun never shines vertically. **Sun rays are always slanting** here resulting in **low temperatures**.
- Because of low temperature, **air compresses and its density increase**. Hence, high pressure is found here. In **northern hemisphere** the belt is called the **North polar high-pressure belt** while it is known as the **South polar high-pressure belt** in the **southern hemisphere**. Winds from these belts blow towards sub-polar low-pressure belts.

While the system of pressure belts provides a structured framework for understanding global atmospheric circulation, it is essential to recognize their dynamic nature. **These belts exhibit seasonal variability, shifting northward in summer and southward in winter in response to the movement of the sun's direct rays between the Tropics of Cancer and Capricorn.** This seasonal shift influences the position of the thermal equator and causes slight variations in the location of pressure belts throughout the year, underscoring the complexity of Earth's atmospheric dynamics.

Morch (Equ.)

Dec/Jan

(Tof calor.)

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June
(Tropic of Cancer)

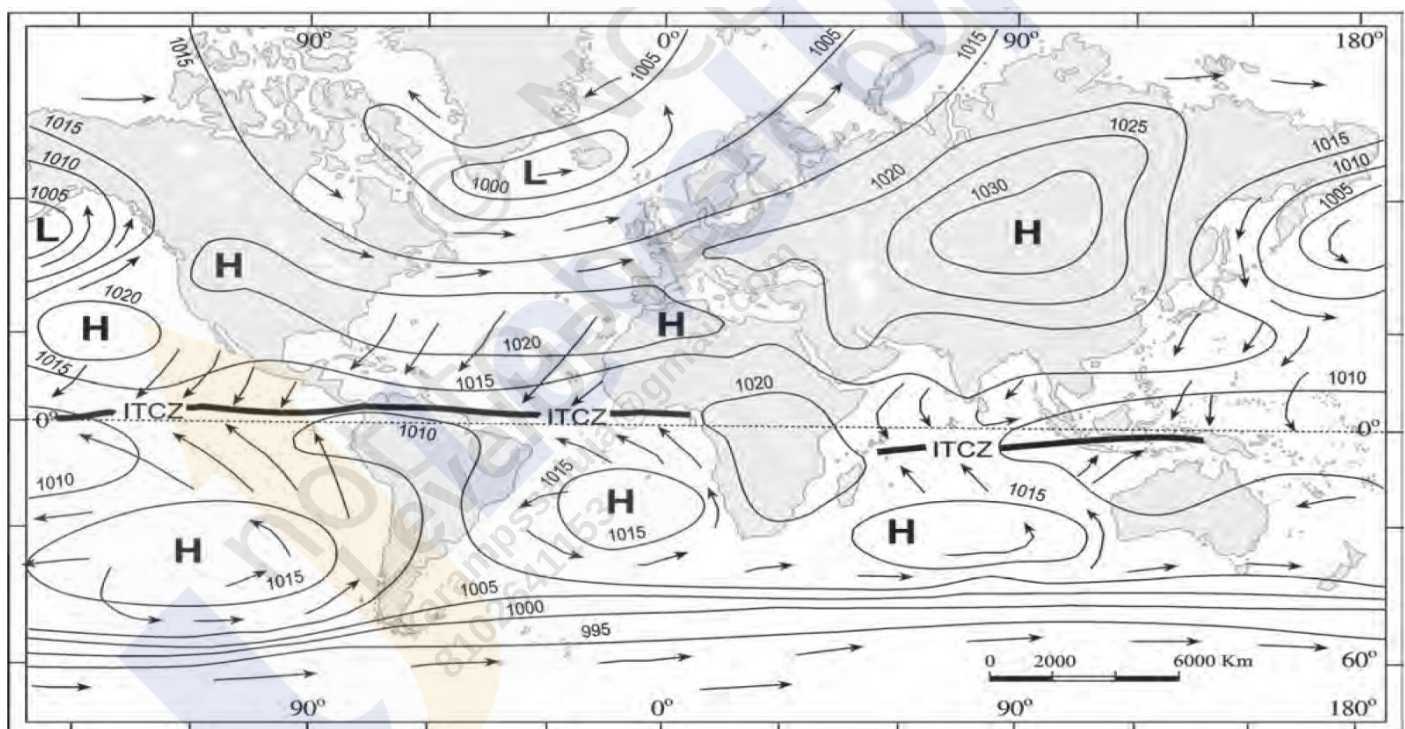
Sept. (Equ.)

Seasonal Distribution of Pressure:

The variation of pressure from place to place and from season to season over the Earth plays an important role in affecting weather and climate. This is why pressure distribution is studied through isobar maps. To avoid the effect of altitude on air pressure, the pressures of all places are reduced to sea level when drawing isobar maps.

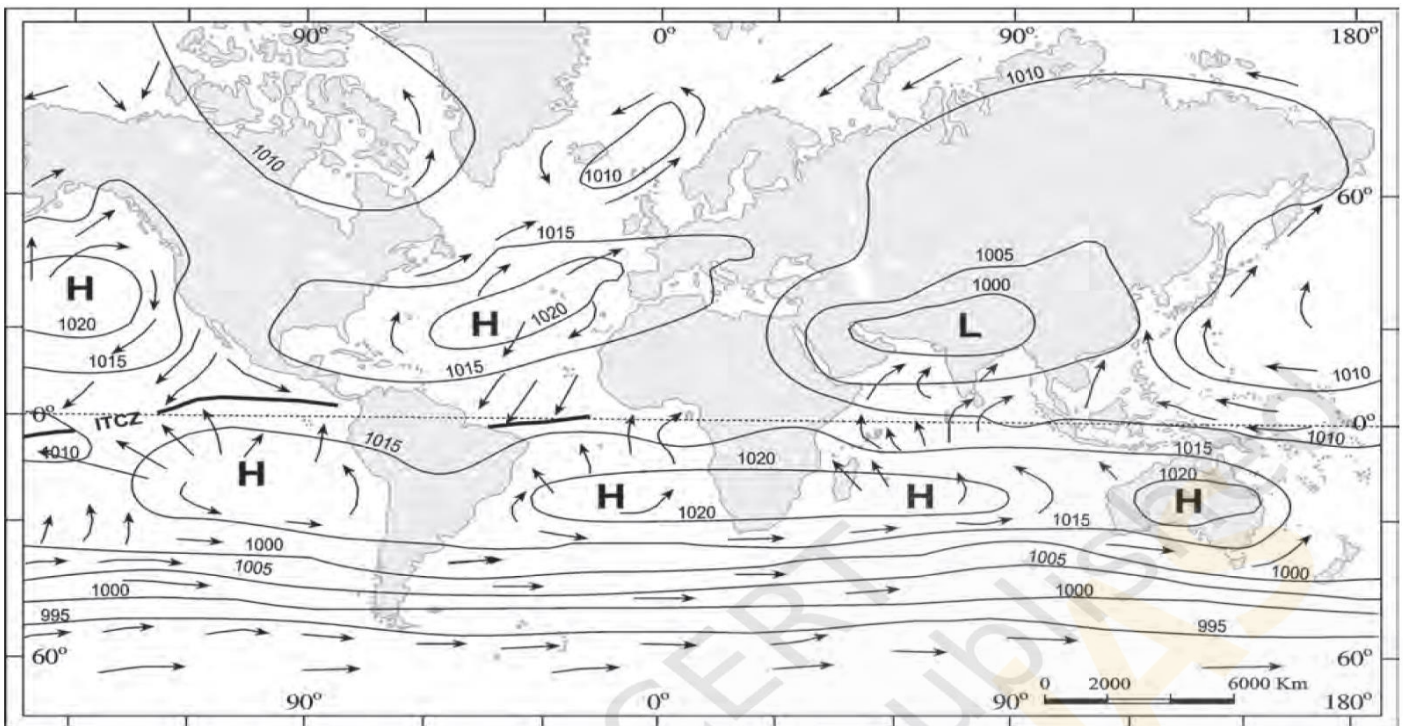
Distribution of Pressure in January:

- In January, with the **southward apparent movement of the Sun**, the **equatorial low-pressure belt shifts slightly south of the mean equatorial position**. Areas of lowest pressure occur in South America, Southern Africa, and Australia due to the land heating up more rapidly than water.
- **Sub-tropical high-pressure cells are centered over the oceans in the southern hemisphere**. The high-pressure belt is interrupted by continental land masses where temperatures are higher, and these cells are well developed in the eastern parts of the oceans dominated by cold ocean currents.
- In the **northern hemisphere, ridges of high pressure occur in the sub-tropical latitudes over the continents**. A well-developed high-pressure cell is found in the interior parts of Eurasia, as the land cools more rapidly than the oceans, resulting in lower winter temperatures compared to surrounding seas.
- In the **southern hemisphere, the sub-polar low-pressure belt circles the Earth** as a continuous belt of low pressure, as there is virtually **no landmass to interrupt** it. In contrast, the northern hemisphere has two low-pressure cells, namely the Icelandic low and the Aleutian low, which develop over the North Atlantic and North Pacific oceans, respectively.



Distribution of Pressure in July:

- In July, the **equatorial low-pressure belt shifts slightly north of the mean equatorial position** due to the **northward apparent movement of the Sun**, causing **all pressure belts to shift northwards**. The **Aleutian and Icelandic lows disappear from the oceans**, while **extensive low-pressure cells develop over landmasses** that had high pressure during winter months.
- In Asia, a significant low-pressure area develops. The **sub-tropical highs of the northern hemisphere are more developed over the Pacific and Atlantic oceans**. In the **southern hemisphere, the sub-tropical high-pressure belt remains continuous**, and the sub-polar low forms a continuous belt, while in the northern hemisphere, there is only a faint oceanic low.



Winds:

Winds are a fundamental component of the Earth's atmospheric system, playing a critical role in shaping weather patterns, climate, and various ecological processes. Defined as the **horizontal movement of air from regions of high pressure to low pressure**, winds arise due to the pressure differences caused by temperature variations, the Earth's rotation, and geographical features. Understanding winds is essential for comprehending the dynamics of the atmosphere, predicting weather changes, and studying climate patterns.

Forces Affecting the Velocity and Direction of Wind

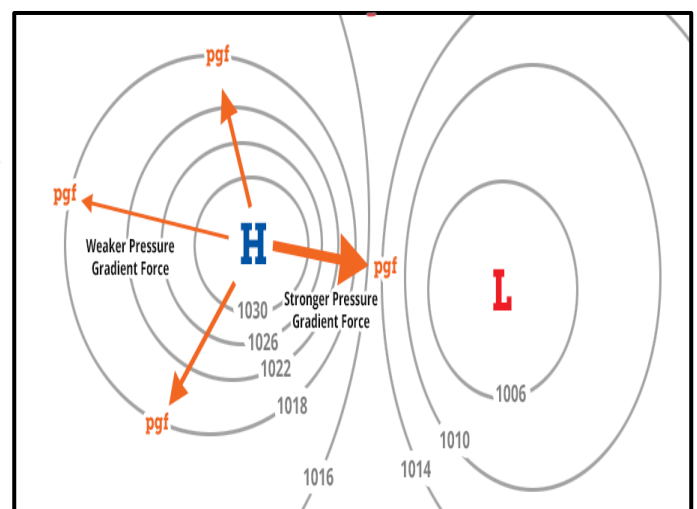
Forces Affecting the Velocity and Direction of Wind:

Pressure Gradient Force (PGF):

The pressure gradient force is the **force that results from differences in atmospheric pressure across the Earth's surface**. This force is the **primary driver of wind**. The direction and magnitude of the pressure gradient ultimately determine wind direction and velocity. **Wind moves from areas of higher pressure to areas of lower pressure**. Consequently, pressure differences cause the wind to blow, and the **greater these differences, the greater the wind velocity**. This interrelationship between gradient force and wind manifests in wind direction and wind speed.

Direction and Speed of Wind:

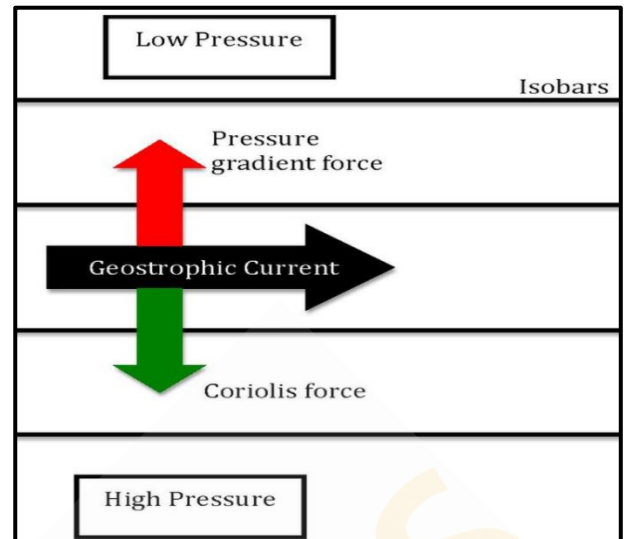
Since wind motion is always from higher pressure towards lower pressure, **the line indicating the direction of wind is always perpendicular to the isobars**. When the **pressure gradient is steep, the wind velocity is higher**, whereas a weak pressure gradient causes the wind to blow at a lower speed. **The spacing of the isobars measures the amount of pressure change occurring over a given distance, expressed as the pressure gradient**. A **steep pressure gradient, indicated by closely spaced isobars, results in high-velocity winds**, while a **weak pressure gradient, indicated by widely spaced isobars, results in light winds**.



Causes of Pressure Differences:

Pressure differences observed on daily weather charts are caused by several factors, with the principal factor being the **unequal heating of the earth's land-sea surface**. Temperature differences generate a pressure gradient, creating winds. Therefore, there is a close relationship between pressure and temperature.

Greater temperature differences result in a steeper pressure gradient and stronger winds. While diurnal variations of temperature and the resultant pressure gradients are confined to a shallow layer of the atmosphere, latitudinal variations in insolation generate atmospheric circulation on a much larger scale.



Coriolis Force:

The Coriolis force, **resulting from the Earth's rotation**, impacts various moving objects, from ocean currents to projectiles. Contrary to its name, it is **not a force per se but an effect influenced by Earth's rotation** and the motion of objects relative to it.

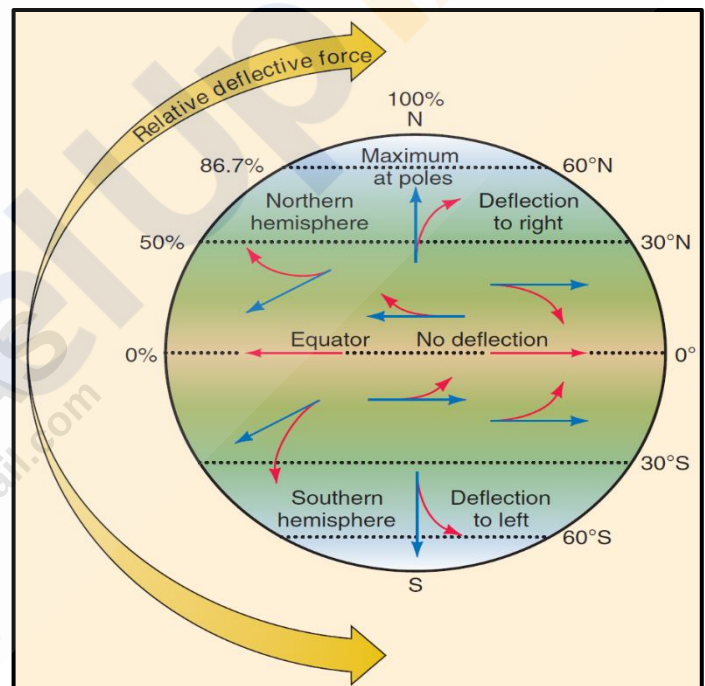
The Coriolis force **varies across latitudes**, being absent at the equator and most pronounced at the poles. It acts **perpendicular to the direction of movement of air**, influencing wind patterns differently in the northern and southern hemispheres.

Directional Influence:

In the **northern hemisphere**, the Coriolis force **deflects moving objects to the right** of their intended path, whereas in the **southern hemisphere**, it deflects them to the **left**. This phenomenon explains the clockwise and counter-clockwise wind circulation around low-pressure centers in these respective hemispheres.

Factors Affecting Coriolis Force:

depends on latitude



The magnitude of the Coriolis force depends on several factors: the horizontal **velocity of the moving object**, its mass, and the **sine of the latitude**. **At higher latitudes**, where this force is **strongest**, the deflection plane aligns parallel to Earth's surface, whereas it **diminishes to zero at the equator**.

Although the **Coriolis force** does not alter the speed of moving objects, it significantly influences their direction by acting perpendicular to their horizontal motion. This force manifests equally in all directions relative to the moving object.

Why do tropical cyclones not form near the equator?

Tropical cyclones, characterized by **intense low-pressure systems** and rotating winds, do not form near the equator due to the **lack of Coriolis force**. The Coriolis force is directly proportional to the angle of latitude. It is maximum at the poles and is absent at the equator.

At the equator (Coriolis force is zero) wind blows perpendicular to the isobars. The **low pressure gets filled instead of getting intensified** i.e., there is no spiralling of air due to zero Coriolis effect. The winds directly get uplifted vertically to form thunderstorms.

Frictional Force:

Frictional force in atmospheric dynamics refers to the **resistance encountered by air as it moves over the Earth's surface, influenced by the roughness and irregularity of the terrain**. This resistance, known as air friction, plays a crucial role in reducing the velocity of wind near the ground, making it **most pronounced at lower altitudes**. As altitude increases, the effect of friction diminishes significantly.

Effects of Surface Characteristics:

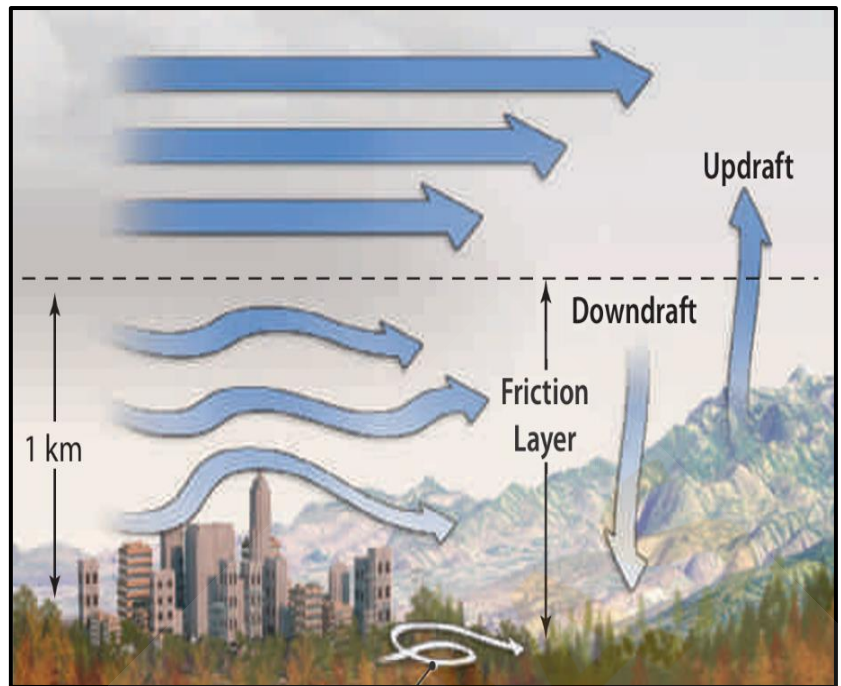
Surface irregularities, such as **hills, valleys, and vegetation**, significantly influence the behavior of wind close to the ground. Within

the **first Kilometer above the surface**, these irregularities cause eddies and turbulence, disrupting the smooth flow of air. **Wind velocity is notably affected, with lower speeds observed near complex terrains.**

Since **Coriolis force is proportionate to wind speed**, the friction consequently reduces it. On the contrary, the **pressure gradient force remains unaffected by wind speed**, hence it **exceeds the Coriolis force along or near the earth's surface**.

Contrast between Land and Sea:

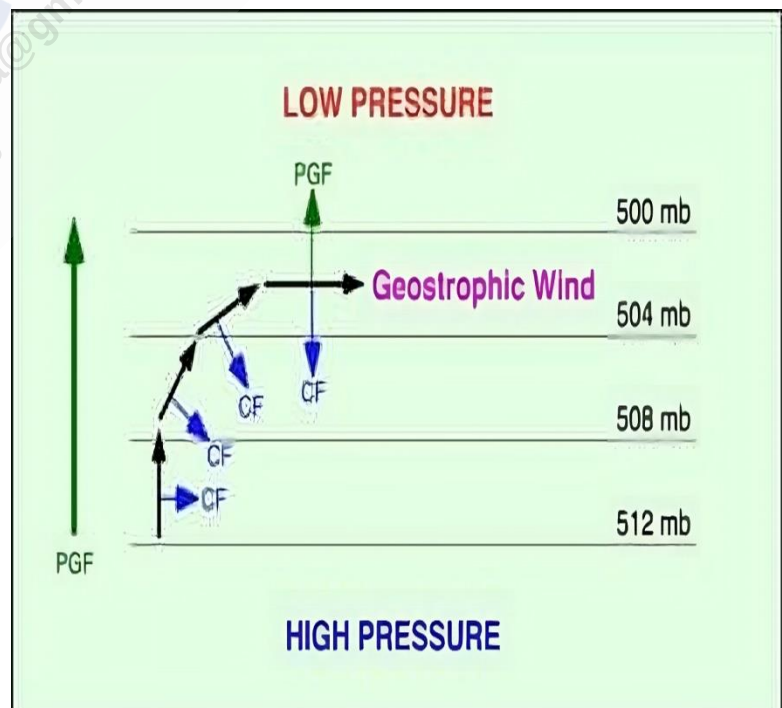
On land, where surface roughness is greater due to diverse topography, **friction exerts a more pronounced effect** on wind velocity. Conversely, **over uniform sea surfaces, friction is minimized**, allowing for **smoother and faster wind flow** at higher altitudes.



Geostrophic Wind:

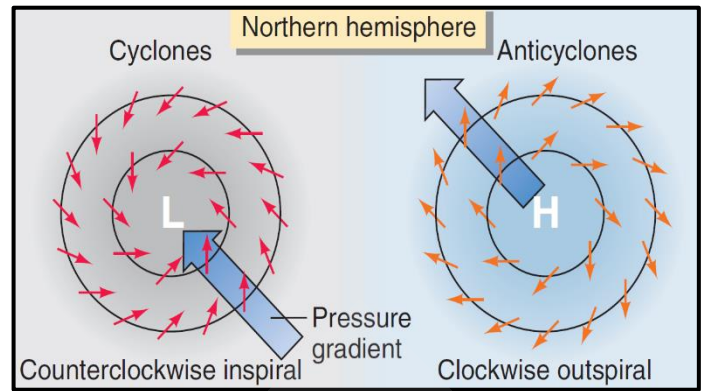
The velocity and direction of wind result from the interaction of various forces. **Winds in the upper atmosphere, typically 2 – 3 km above the Earth's surface, are influenced primarily by the pressure gradient and the Coriolis force**. These winds are largely **free from surface frictional effects**.

When isobars (lines of equal pressure) are straight and there is **minimal friction**, the **pressure gradient force balances with the Coriolis force**. As a result, the **wind blows parallel to the isobars**. This specific wind pattern is known as the '**Geostrophic wind**.' In this scenario, the wind's movement is predictable and largely influenced by the surrounding pressure systems.



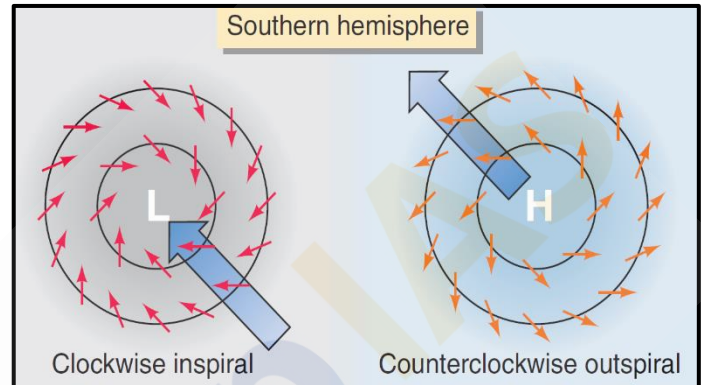
Circulation Patterns - Cyclonic and Anticyclonic:

The movement of wind **around low-pressure** systems exhibits **cyclonic circulation**, where **air converges and rises**. Conversely, **around high-pressure systems**, known as **anticyclones**, **air subsides** from above and **diverges at the surface**. The direction of these winds varies depending on the hemisphere, influencing weather patterns and atmospheric circulation.



Surface and Upper-Level Interactions:

Wind circulation at the Earth's surface around highs and lows often correlates with patterns observed in the upper atmosphere. Convergence and divergence of air play crucial roles, influenced by factors such as eddies, convection currents, orographic uplift, and uplift along frontal boundaries. These processes are integral to cloud formation and precipitation, shaping regional climates and weather phenomena.



Type of Winds:

Throughout history, humans have noted distinct wind patterns across the Earth. In some regions, winds consistently blow from one direction year-round, while in others, wind direction shifts with the seasons. There are also areas where winds are highly unpredictable, lacking any discernible pattern.

Despite these variations, winds are generally classified into three categories:

- Primary Winds or Prevailing Winds or Permanent Winds or Planetary Winds
- Secondary Winds or Periodic Winds
- Tertiary Winds or Local Winds.

Primary Winds or Permanent Winds or Planetary Winds:

Planetary winds, also known as **Prevailing Winds**, are the large-scale winds that **blow steadily from high-pressure regions to low-pressure regions across the Earth's surface**. They cover extensive areas over continents and oceans. They are easterlies and westerlies and polar easterlies.

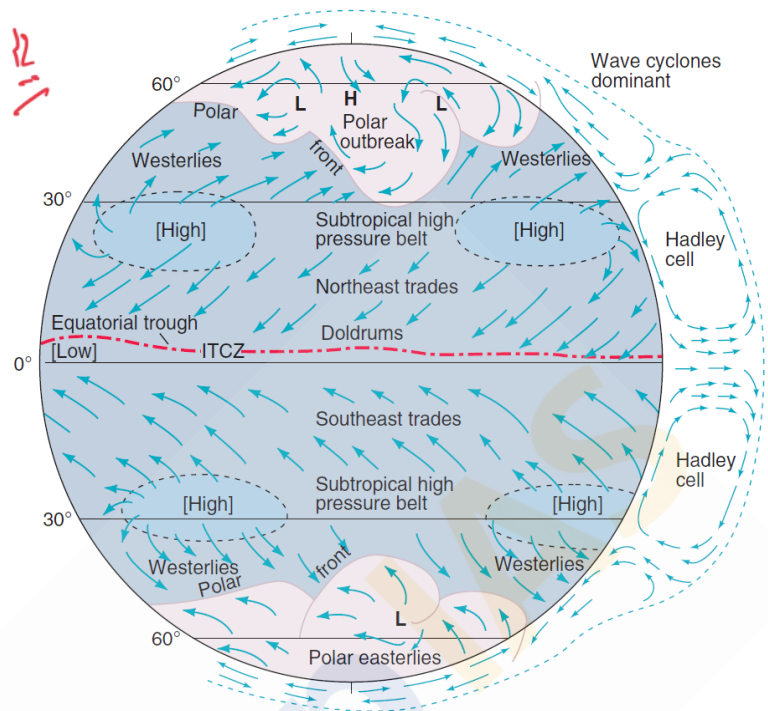
The Easterlies or Trade Winds:

- These are winds blowing **from the subtropical high-pressure areas towards the equatorial low-pressure areas**. They are confined to latitudes between **30°N and 30°S across the Earth's surface**.
- In the **Northern hemisphere**, they are known as **northeast trades**, while in the **Southern hemisphere**, they are called **southeast trades**. These names derive from their consistent direction, which **historically aided sea merchants in navigation**.
- The deflection of trade winds from their ideal north-south path is influenced by the Coriolis force and Ferrel's law. In the **northern hemisphere, they deflect to the right**, and in the **southern hemisphere, to the left**.

- Originating from stable, descending air masses in the subtropical high-pressure belt, trade winds become warm and humid as they approach the equator, leading to **heavy rainfall due to convergence at the equatorial zone**. The **eastern parts, associated with cool ocean currents**, tend to be **drier and more stable** compared to their western counterparts.

The Westerlies:

- The westerlies are winds blowing from the **subtropical high-pressure belts (30°-35°)** towards the **sub-polar low-pressure belts (60°-65°)** in both hemispheres. These winds move from **southwest to northeast** in the **Northern hemisphere** and from **northwest to southeast** in the **Southern hemisphere**.
- In the **southern hemisphere**, the westerlies are **stronger and more persistent** due to the **vast expanse of water**. Conversely, in the **northern hemisphere**, they are **irregular** because of the uneven relief of extensive landmasses. The dominance of land in the northern hemisphere makes the westerlies more complex, less effective during the summer, and more vigorous in the winter.
- The westerlies bring **significant precipitation to the western parts of continents**, such as the northwest European coasts, as they pick up substantial moisture while passing over vast stretches of the oceans.
- The westerlies become more **vigorous in the southern hemisphere** due to the dominance of oceans and lack of land. Their velocity increases southward, turning stormy and associated with boisterous gales. They are best developed between 40° and 65°S latitudes, known as the **Roaring Forties, Furious Fifties, and Shrieking Sixties** — dreaded terms for sailors.
- The poleward boundary of the westerlies fluctuates significantly with many seasonal and short-term changes. These winds produce wet spells and variability in weather conditions.



Polar Easterlies:

- Polar easterlies blow from polar regions towards sub-polar low-pressure regions. They are dry, cold prevailing winds blowing from **north-east to south-west direction** in **Northern hemisphere** and **south-east to the north-west** in **Southern hemisphere**.
- They blow from the **polar high-pressure areas of the sub-polar lows**.

Secondary Winds or Periodic Winds:

Secondary winds are also known as periodic winds. Unlike planetary winds, **the pattern of wind circulation in secondary winds are modified in different seasons** due to the shifting of regions of maximum heating, pressure and wind belts.

In other words, they **change their direction with the seasons**.

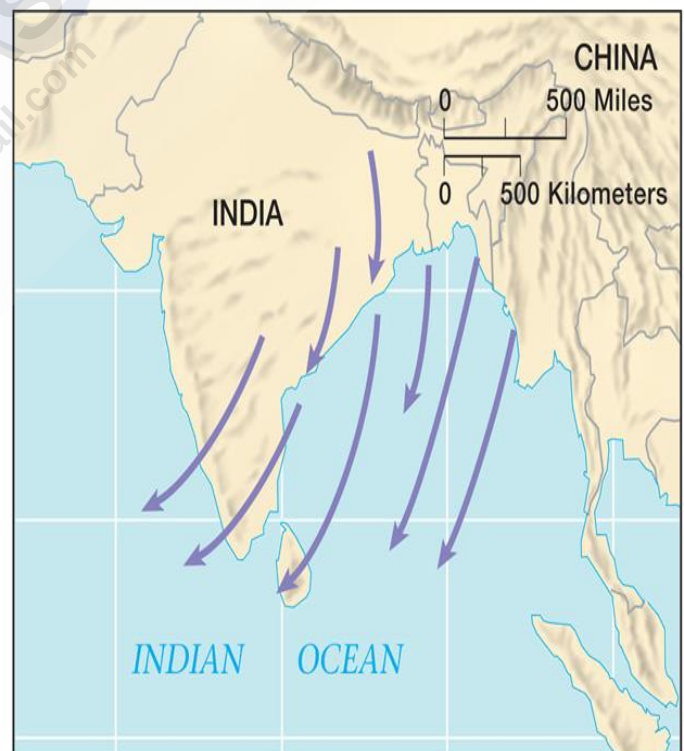
The most pronounced effect of such a shift is noticed in the **monsoons**, especially over southeast Asia. Other examples of periodic winds include **land and sea breezes, mountain and valley breezes, cyclones, anticyclones, and air masses**.

Monsoon Winds:

- The word '**Monsoon**' has been derived from the Arabic word '**Mausim**' meaning season. The winds that reverse their direction with the change of seasons are called monsoon winds. Monsoons have traditionally been explained as **large-scale land and sea breezes**, viewed as a convectional circulation on a grand scale. But this explanation does not hold good now. Now a days the monsoon is generally accepted as **seasonal modification of the general planetary wind system**.
- A defining characteristic of monsoons is the seasonal reversal of wind direction. During the **summer**, the southern hemisphere's trade winds are drawn northward by the apparent **northward movement of the sun** and a powerful **low-pressure** core in the northwest of the **Indian subcontinent**. As these **winds cross the equator**, they are **deflected** to the right due to the **Coriolis force**, becoming southwest monsoons. After traveling a considerable distance **over the ocean**, these winds become **saturated** with moisture and bring heavy rainfall to India and its neighboring regions.
- In **winter**, this process is reversed. A **high-pressure** core forms **north of the Indian subcontinent**, generating divergent winds through **anticyclonic movement**. These winds travel southwards towards the equator, a movement enhanced by the **apparent southward shift of the sun**. Known as **northeast or winter monsoons**, these winds cause some **precipitation along India's east coast**.
- Monsoon winds influence a vast region, including **India, Pakistan, Bangladesh, Myanmar (Burma), Sri Lanka, the Arabian Sea, Bay of Bengal, southeastern Asia, northern Australia, and China**. In eastern Asiatic countries such as China and Japan, the winter monsoon is more powerful than the summer monsoon. We will explore monsoons in greater detail when studying the Indian climate.



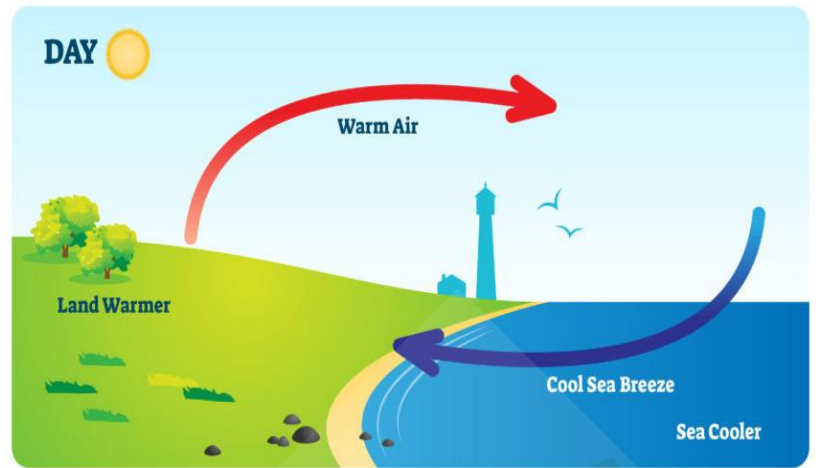
Summer



Winter

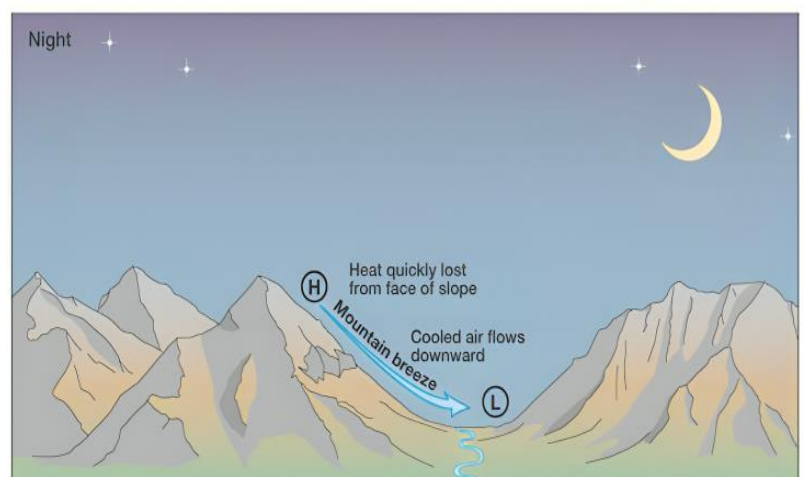
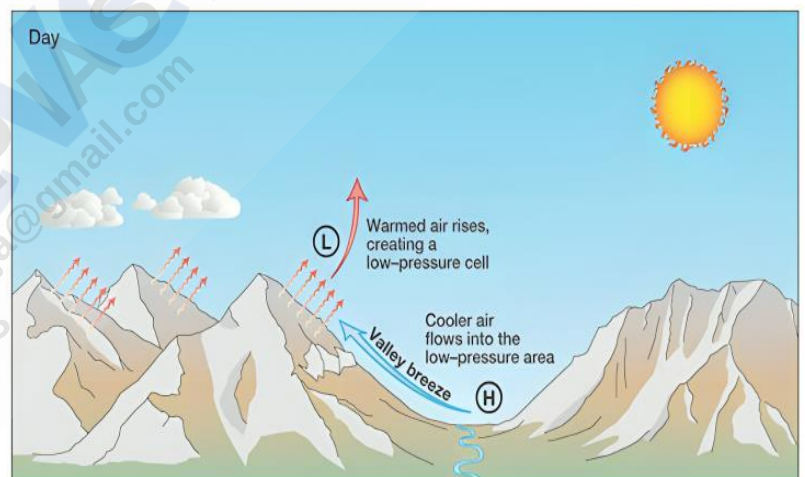
Land Breeze and Sea Breeze:

- The land and sea absorb and transfer heat differently, leading to variations in temperature and pressure that create breezes. During the **day**, the **land heats up faster and becomes warmer than the sea**. As a result, the **air over the land rises**, creating a **low-pressure area**. In contrast, the sea **remains relatively cool**, and the pressure over the sea is relatively high. This difference in pressure, known as the **pressure gradient**, **causes wind to blow from the sea to the land**, forming a **sea breeze**.
- At **night**, the conditions reverse. The **land loses heat faster than the sea** and becomes cooler. Consequently, **the pressure gradient shifts**, with higher pressure over the cooler land and lower pressure over the warmer sea. This shift causes **wind to blow from the land to the sea**, resulting in a **land breeze**.



Valley Breeze and Mountain Breeze:

- In mountainous regions, distinct wind patterns develop due to the temperature differences between slopes and valleys. During the **day**, the **slopes of mountains get heated** by the sun, causing the **air in these areas to warm up** and rise. To fill the resulting gap, **cooler air from the valley blows upslope**, creating what is known as a **valley breeze**.
- At **night**, the process reverses. The **slopes lose heat rapidly and cool down**, causing the air to become denser. This **denser, cooler air descends** into the valley, forming a **mountain breeze**. The flow of air from the high plateaus and ice fields into the valley at night is also referred to as **katabatic wind**.
- The valley and mountain breezes are also named as **anabatic** and **katabatic breezes** respectively.



Katabatic Wind on Leeward Side:

- Another type of **warm wind (katabatic wind)** occurs on the **leeward side** of mountain ranges but in a different context.
- As **moist winds cross the mountain ranges**, they condense and precipitate. When these winds **descend down** the leeward side, the now dry air undergoes **warming due to the adiabatic process**. This warming can be so significant that it can rapidly melt snow on the slopes. These warm, descending winds are another manifestation of katabatic winds, highlighting their diverse impacts in mountainous regions.

Tertiary Winds or Local Winds:

- Local winds, also known as tertiary winds, arise due to **variations in temperature and pressure** within a specific region. These winds are typically confined to the lowest levels of the troposphere and have a **limited geographical extent**.
- Local winds are generated by differences in temperature and pressure within a localized area. These variations can be due to several factors, including the **uneven heating of the Earth's surface**, **topographical features**, and the **presence of water bodies**.

Hot Local Winds:

Loo:

- Loo are hot and dry winds that blow strongly over the **northern plains of India and Pakistan** during **May and June**. These winds move from **west to east** and are usually experienced in the **afternoons**. Their temperature ranges between **45°C to 50°C**.

Foehn:

- Foehn is a **strong, dusty, dry, and warm local wind** that develops on the **leeward side of the Alps Mountain ranges**. A regional pressure gradient forces the air to ascend and cross the barrier, **causing precipitation on the windward side**. After crossing the mountain crest, the Foehn winds descend on the leeward side as warm and dry winds. The temperature of the Foehn winds varies from **15°C to 20°C**, which helps in **melting snow, making pasture land ready for animal grazing**, and helping grapes to ripen early.

Chinook:

- Chinook is the name of **hot and dry local winds** that move down the **eastern slopes of the Rockies in the U.S.A. and Canada**. The term "**Chinook**" means '**snow eater**' as these winds help in **melting snow early**. They keep the **grasslands clear of snow**, which is beneficial for ranchers.

The Santa Ana:

- The Santa Ana is a **hot, dry, strong, blustery, föhn-type wind** that blows from the north-east or east over **southern California**, carrying **large quantities of dust**. It is most frequent in winter but can also occur in spring or autumn. These winds can cause considerable damage, drying out vegetation and increasing wildfire risks. Once fires start, the winds fan the flames and hasten the spread of fires. In spring, Santa Ana winds can **cause damage to fruit trees**.

Sirocco:

- Sirocco is a **Mediterranean wind** that **comes from the Sahara** and reaches hurricane speeds in North Africa and Southern Europe. It arises from a **warm, dry, tropical air mass** pulled northward by low-pressure cells moving eastward across the Mediterranean Sea. The Sirocco **causes dusty dry conditions** along the northern coast of Africa, storms in the Mediterranean Sea, and **cool wet weather in Europe**.

Harmattan:

- Harmattan are **warm and dry winds** blowing from the north-east and east to west in the **eastern parts of the Sahara Desert**. These winds become **extremely dry** due to their journey over the Sahara Desert, picking up sand along the way. In the western coast of Africa, the arrival of Harmattan reduces humidity, **making the weather dry and pleasant**. This is why harmattan is **known as the 'doctor' in the Guinea coastal area**. It is very **dusty and stormy**, often associated with dust storms that reduce visibility. It is a special type of north-east trade wind, becoming extremely warm due to the hot and dry Sahara Desert.

The Khamsin:

- The Khamsin is a **hot, dry, dust-laden** southerly wind over **Egypt, the Red Sea**, and eastern parts of the **Mediterranean Sea** ahead of eastward-moving depressions. Occurring between **February and June**, it is most frequent in March and April. **The Khamsin is a Sirocco wind**, and its name comes from the Arabic word for '**fifty**', referring to the approximate length of time the wind blows. In the nineteenth century, the plague in Egypt was at its worst when the Khamsin was blowing.

Cold Local Winds:**Mistral:**

- The mistral is a prevalent local cold wind originating from the **Alps** and sweeping across **France** towards the **Mediterranean Sea** through the **Rhone valley**. Known for its **high velocity and dryness**, mistral winds can bring **temperatures below freezing**, prompting locals to protect orchards and gardens with dense hedges and orient their homes towards the Mediterranean Sea.

Bora:

- Another significant cold wind in **eastern Europe**, the bora, descends from the **Dinaric Alps** to the **Adriatic Coast**. This **blustery wind**, blowing from the north or northeast at speeds exceeding **100 mph (160 kph)**, influences weather patterns in the region during winter.

Papagayo:

- The papagayo is a strong northeasterly wind affecting the **Pacific coast of Central America**, from **Guatemala** to **Costa Rica's Gulf of Papagayo**. Originating from a cold air mass descending through Central American mountains, it brings cold, clear, and blustery weather conditions.

Blizzard:

- Blizzards are **violent storms** characterized by **cold, powdery polar winds** laden with **dry snow**. Common in the **Polar Regions, Siberia, Canada**, and the **USA**, these winds bring low visibility due to snow and ice crystals. Temperatures drop suddenly **below freezing**, accompanied by **thick snow cover and cold waves**, reaching southern

USA states due to the absence of east-west mountain barriers. Known as '**norther**' in the **southern USA** and '**burran**' in **Siberia**, blizzards are impactful weather events.

Pampero:

- In **Argentina and Uruguay**, particularly around the **Rio de la Plata area**, pampero refers to severe line squalls associated with strong cold fronts. These events are marked by **rain, thunder, lightning, a sharp temperature drop**, and a **sudden wind direction shift** from northerly or north-westerly to southerly or south-westerly. Pamperos are most likely to occur between **June and September**.

Buran:

- It's a cold, sometimes quite strong wind that blows through the **steppes** of the **Sarmatic Plain, west of the Urals**. They take up dual forms - one during winter and the other in summer. In the summer, the Buran is a hot, dry wind that stirs up sandstorms. In the winter, it is severely cold and frequently accompanied by blizzards.

Levanter:

- It is a mighty easterly wind of the **Mediterranean**, particularly in the **Strait of Gibraltar**. They are accompanied by cloudy, foggy, and sometimes rainy weather, especially in winters.

